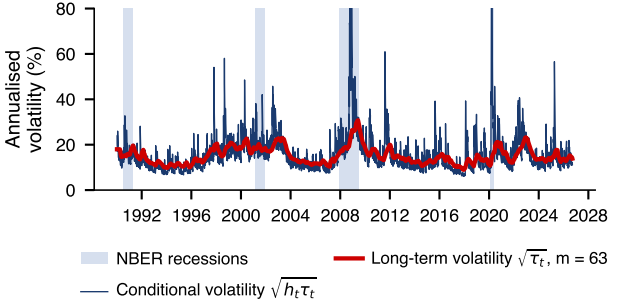
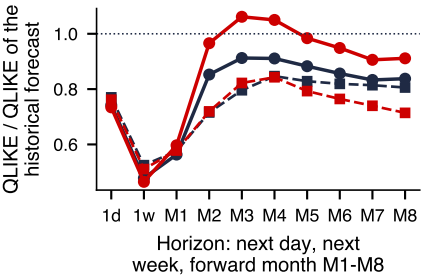

Annualised volatility (%)
80
60
40
20
0
1992
1996
2000
2004
2008
2012
2016
2020
2024
2028
NBER recessions
Long-term volatility $\sqrt{\tau_t}$, m = 63
Conditional volatility $\sqrt{h_t \tau_t}$


QLIKE / QLIKE of the historical forecast
1.0
0.8
0.6
1d
1w
M1
M2
M3
M4
M5
M6
M7
M8
Horizon: next day, next week, forward month M1-M8
GJR-GARCH, 2010-2026
GJR-GARCH, 2010-2019
MF2-GARCH, 2010-2026
MF2-GARCH, 2010-2019